Supplementary Materials

Rethinking Parameter Invariance in Cognitive Modeling

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S1 Formal derivations

The main text presents the scientific argument and primary simulation results. This supplement gives the supporting scalar derivations and reports additional analyses needed to interpret the simulations. Numerical summaries, fit diagnostics, and replication-level results accompany the code.

S1.1 Loss scale and assumptions

For $T \in \{A, B\}$, let L_T denote the population effective loss under the declared nuisance policy, and let $\theta_T^* = \arg\min_\theta L_T(\theta)$. For a scalar focal parameter, the local population excess loss on Task T is

$$L_T(\theta) - L_T(\theta_T^*) \approx \frac{w_T}{2}(\theta - \theta_T^*)^2, \qquad w_T > 0. \tag{S1}$$

Except where a result is identified as exact, the scalar and matrix results below are derived within the local quadratic representation and are only approximate for the original loss. Scalar results use equal external task weights and a common predictive tolerance ε except where noted. Unless otherwise stated, transfer carries the same focal value across tasks on a common scale, adequacy is defined solely by the declared effective loss, and the relevant global minima are assumed to be attained. Statements involving boundary ratios assume a positive tolerance. All losses, curvatures, and tolerances are interpreted on a common scale within each derivation. Multiplying a loss by a positive constant multiplies both its curvature and tolerance by that constant and leaves the corresponding acceptable region unchanged. The sampling derivation below uses total negative log likelihood, whereas the simulation checks use mean per-observation loss. The simulation tolerances are $\varepsilon = 0.02$ for Simulation 1 and $\varepsilon = 0.007$ nats per trial for Simulation 2.

S1.2 Joint compromise under general task weights

First allow the joint population objective to use positive external task weights ν_A and ν_B. Writing

$$W = \nu_A w_A + \nu_B w_B,$$

direct differentiation gives

$$\theta_{\text{joint}}^* = \frac{\nu_A w_A \theta_A^* + \nu_B w_B \theta_B^*}{W}. \tag{S2}$$

If $\delta = \theta_B^* - \theta_A^*$, the excess loss of this value on Task B is

$$\Delta L_B^{\text{joint}} = \frac{w_B}{2}\left(\frac{\nu_A w_A}{W}\right)^2 \delta^2, \tag{S3}$$

with the analogous expression obtained by exchanging A and B. The target curvature outside the squared term remains w_B: the external weights select the joint value, but Task B's loss evaluates it.

Writing $a = \nu_A w_A / W$, the joint value lies a fraction $1 - a$ of the discrepancy from θ_A^* and a fraction a from θ_B^*. Under a common tolerance with acceptable radii $r_T = \sqrt{2\varepsilon / w_T}$, the joint criterion therefore passes when

$$|\delta| \leq \min\left\{\frac{r_A}{1-a}, \frac{r_B}{a}\right\}. \tag{S4}$$

Because both denominators are at most one, this boundary is never smaller than the boundary $\min(r_A, r_B)$ for bidirectional transfer, as the exchange inequality in Section S1.9 requires. With equal curvature, the ratio of the two boundaries is $1/\max(a, 1-a)$. It equals 2 under equal external task weights, falls to 1.25 when one task receives four times the weight of the other, and approaches 1 as one task dominates the objective. The factors reported in the main text therefore depend on equal external task weights, whereas the ordering of the boundaries does not.

The central scalar results use $\nu_A = \nu_B = 1$. In that case, let $\delta = \theta_B^* - \theta_A^*$ and write $W_0 = w_A + w_B$. The joint optimum and its displacements from the two task-specific optima then follow.

$$\theta_{\text{joint}}^* = \frac{w_A \theta_A^* + w_B \theta_B^*}{W_0}, \tag{S5}$$

$$\theta_{\text{joint}}^* - \theta_A^* = \frac{w_B}{W_0}\delta, \qquad \theta_{\text{joint}}^* - \theta_B^* = -\frac{w_A}{W_0}\delta. \tag{S6}$$

Substitution carries a task-specific optimum across the full discrepancy, whereas the joint value incurs only the corresponding weighted displacement. Within the quadratic approximation, the two joint excess losses are therefore

$$\Delta L_A^{\text{joint}} = \frac{w_A}{2}\left(\frac{w_B}{W_0}\right)^2 \delta^2, \qquad \Delta L_B^{\text{joint}} = \frac{w_B}{2}\left(\frac{w_A}{W_0}\right)^2 \delta^2. \tag{S7}$$

S1.3 Directional and bidirectional boundaries

For A→B transfer and its reverse, the quadratic substitution losses are $w_B\delta^2/2$ and $w_A\delta^2/2$. Defining c as a boundary on the squared discrepancy δ^2, the two directional boundaries under the common tolerance are

$$c_{A\to B} = \frac{2\varepsilon}{w_B}, \qquad c_{B\to A} = \frac{2\varepsilon}{w_A}. \tag{S8}$$

Under the same tolerance, Equation S7 shows that the Task B component of the joint criterion passes when $\delta^2 \leq c_{\text{joint},B} = c_{A\to B}(W_0/w_A)^2 = c_{A\to B}(1 + w_B/w_A)^2$. This boundary exceeds the A→B transfer boundary by the same factor by which the jointly selected value reduces Task B's excess loss. With equal curvature, it is four times as large on the squared scale and twice as large on the original parameter scale.

Bidirectional transfer under substitution requires both directions to pass, so

$$c_{\text{sub}}^{\text{bi}} = \frac{2\varepsilon}{\max(w_A, w_B)}. \tag{S9}$$

Applying the same tolerance to both components of the joint criterion and using Equation S7 gives

$$c_{\text{joint}}^{\text{bi}} = \min\left\{\frac{2\varepsilon W_0^2}{w_A w_B^2}, \frac{2\varepsilon W_0^2}{w_B w_A^2}\right\}. \tag{S10}$$

For example, if $w_A \geq w_B$, the second joint component is the more restrictive one and $c_{\text{sub}}^{\text{bi}} = 2\varepsilon/w_A$. Their ratio is then $W_0^2/(w_A w_B)$; exchanging A and B gives the same result for the other ordering. Thus

$$\frac{c_{\text{joint}}^{\text{bi}}}{c_{\text{sub}}^{\text{bi}}} = \frac{(w_A + w_B)^2}{w_A w_B}. \tag{S11}$$

Each squared-discrepancy boundary in this subsection is proportional to ε. Changing the common tolerance therefore moves every boundary in proportion to $\sqrt{\varepsilon}$ on the original parameter scale but leaves their ratios unchanged, so a range in which joint adequacy holds while bidirectional transfer fails exists at every positive tolerance. Section S3 reports the corresponding check for the nonlinear psychometric model.

S1.4 Calibration

The Task B component of the joint criterion can be calibrated to the A→B transfer boundary under substitution by setting

$$\varepsilon_{\text{joint},B}^{\text{cal}} = \varepsilon \left(\frac{w_A}{W_0}\right)^2. \tag{S12}$$

Direct substitution shows that

$$\Delta L_B^{\text{joint}} \leq \varepsilon_{\text{joint},B}^{\text{cal}} \quad \Longleftrightarrow \quad \frac{w_B}{2}\delta^2 \leq \varepsilon \quad \Longleftrightarrow \quad \delta^2 \leq c_{A\to B}. \tag{S13}$$

The reverse calibration exchanges A and B. A single tolerance that aligns the maximum-loss joint criterion with bidirectional transfer under substitution is

$$\varepsilon_{\text{joint}}^{\text{cal}} = \varepsilon \frac{w_A w_B}{(w_A + w_B)^2}. \tag{S14}$$

This calibration matches the selected population boundary. It does not make the jointly selected value a source-selected value or align the procedures' finite-sample behavior.

S1.5 Shared-versus-separate model comparison

On a common total negative-log-likelihood scale, the population cost of replacing the two task-specific optima with the shared compromise is the sum of the two losses in Equation S7:

$$\Delta L_{\text{total}} = \Delta L_A^{\text{joint}} + \Delta L_B^{\text{joint}} \tag{S15}$$

$$= \frac{1}{2W_0^2}\left(w_A w_B^2 + w_B w_A^2\right)\delta^2 = \frac{w_A w_B}{2(w_A + w_B)}\delta^2. \tag{S16}$$

If the complexity cost for freeing one parameter is κ, the corresponding expected-fit boundary is

$$\delta^2 \leq \frac{2\kappa(w_A + w_B)}{w_A w_B}. \tag{S17}$$

This expected-fit comparison treats the loss difference as deterministic and does not include sampling fluctuation. Standard penalties such as those of AIC and BIC apply to the ordinary likelihood, so externally weighted objectives require their own justification. With equal curvature

w, the cost in Equation S16 is $w\delta^2/4$, half the directional transfer loss $w\delta^2/2$, whereas the shared compromise incurs $w\delta^2/8$ on each task, one quarter of that transfer loss. Comparing Equation S17 with $c_{A\to B}$ shows that, on a common total-loss scale, discrepancies at which a criterion with penalty κ favors the shared model while A→B transfer fails exist exactly when $\varepsilon < \kappa(1 + w_B/w_A)$. When $\varepsilon > \kappa(1 + w_B/w_A)$, the criterion instead favors separate parameters for some discrepancies at which A→B transfer still succeeds. With equal curvature, these conditions become $\varepsilon < 2\kappa$ and $\varepsilon > 2\kappa$. Model comparison evaluates this aggregate cost against its inferential or complexity criterion. Transfer instead evaluates a source-selected value in a particular target task, which is why the two decisions need not coincide.

S1.6 Coincident optima and same-sample optimism

The sampling benchmark assumes standard regular likelihood conditions, approximately unbiased scores, information-matrix equality, equal external task weights, independent task datasets, and coincident population optima. Let $d = \theta - \theta^*$ and $g_T = \partial \widehat{L}_T(\theta^*)/\partial\theta$. The local empirical loss difference from the population optimum for Task T is

$$\widehat{L}_T(\theta) - \widehat{L}_T(\theta^*) \approx g_T d + \frac{w_T}{2} d^2, \qquad \mathbb{E}(g_T) \approx 0, \qquad \mathrm{Var}(g_T) \approx w_T. \tag{S18}$$

With independent task scores, the displacement of the joint estimator is

$$d_{\text{joint}} \approx -\frac{g_A + g_B}{w_A + w_B}.$$

The two expectations needed below are

$$\mathbb{E}(g_B d_{\text{joint}}) \approx -\frac{w_B}{w_A + w_B}, \qquad \mathbb{E}(d_{\text{joint}}^2) \approx \frac{1}{w_A + w_B}. \tag{S19}$$

Its expected population excess loss on Task B, equivalently its expected loss difference on independent validation data with the same Task B design and total-loss normalization, is

$$\mathbb{E}(\Delta L_{B,\text{fresh}}^{\text{joint}}) \approx \frac{w_B}{2(w_A + w_B)}. \tag{S20}$$

Using the same Task B observations for fitting and scoring instead gives

$$\begin{aligned} \mathbb{E}(\Delta \widehat{L}_{B,\text{same}}^{\text{joint}}) &\approx \mathbb{E}\left[g_B d_{\text{joint}} + \frac{w_B}{2} d_{\text{joint}}^2 \right] \\ &\approx -\frac{w_B}{2(w_A + w_B)}. \end{aligned} \tag{S21}$$

Here $\Delta \widehat{L}_{B,\text{same}}^{\text{joint}}$ is the empirical loss difference relative to the population optimum θ^*, not a nonnegative excess loss relative to Task B's empirical sample optimum. The negative expectation is therefore consistent with the local same-sample optimism described in the main text. It is separate from the pooling advantage: a source estimate from Task A has fresh Task B expected excess loss approximately $w_B/(2w_A)$ under the same assumptions.

S1.7 Profiling nuisance parameters

Assume a twice-differentiable loss and an interior optimum. Let d_θ and $\mathbf{d}_\phi$ denote local focal and nuisance displacements, and partition the Hessian of the unprofiled target loss $\mathbb{E}[\mathcal{L}_B(\theta, \phi; D_B)]$ with respect to (θ, ϕ), evaluated at its unconstrained population optimum, as

$$\mathbf{H}_B^{\text{full}} = \begin{pmatrix} H_{\theta\theta} & \mathbf{H}_{\theta\phi} \\ \mathbf{H}_{\phi\theta} & \mathbf{H}_{\phi\phi} \end{pmatrix}.$$

When $\mathbf{H}_{\phi\phi}$ is positive definite, minimizing the quadratic loss over the nuisance displacement gives

$$\mathbf{d}_\phi^*(d_\theta) = -\mathbf{H}_{\phi\phi}^{-1}\mathbf{H}_{\phi\theta} d_\theta,$$

so the profile curvature for the focal parameter is the Schur complement

$$H_{\theta|\phi} = H_{\theta\theta} - \mathbf{H}_{\theta\phi}\mathbf{H}_{\phi\phi}^{-1}\mathbf{H}_{\phi\theta}. \tag{S22}$$

This Schur complement is the curvature of the effective loss used in the main text when the nuisance parameters are re-estimated. For one nuisance parameter, define

$$\gamma_B = \frac{H_{\theta\phi}}{\sqrt{H_{\theta\theta} H_{\phi\phi}}}.$$

Positive definiteness of $\mathbf{H}_B^{\text{full}}$ implies $|\gamma_B| < 1$, and Equation S22 becomes

$$H_{\theta|\phi} = H_{\theta\theta}(1 - \gamma_B^2).$$

Because the local acceptable radius varies as $H_{\theta|\phi}^{-1/2}$, allowing ϕ to adapt multiplies that radius by $(1 - \gamma_B^2)^{-1/2}$ at the same positive predictive tolerance, relative to holding ϕ at its value in the unconstrained population optimum. This calculation makes explicit why the declared nuisance policy changes the effective transfer criterion.

S1.8 A multivariate focal claim

Let $\boldsymbol{\theta}$ be a focal vector, let $\boldsymbol{\delta} = \boldsymbol{\theta}_B^* - \boldsymbol{\theta}_A^*$, and assume the profiled curvature matrices $\mathbf{W}_A$ and $\mathbf{W}_B$ are positive definite. Define

$$\mathbf{D} = \nu_A \mathbf{W}_A + \nu_B \mathbf{W}_B.$$

The local joint optimum is

$$\boldsymbol{\theta}_{\text{joint}}^* = \mathbf{D}^{-1} \left(\nu_A \mathbf{W}_A \boldsymbol{\theta}_A^* + \nu_B \mathbf{W}_B \boldsymbol{\theta}_B^* \right). \tag{S23}$$

With $\mathbf{M} = \mathbf{D}^{-1}\nu_A \mathbf{W}_A$, its Task B excess loss is

$$\Delta L_B^{\text{joint}} = \frac{1}{2} \boldsymbol{\delta}^\top \mathbf{M}^\top \mathbf{W}_B \mathbf{M} \boldsymbol{\delta}, \tag{S24}$$

whereas transferring Task A's optimum incurs $\Delta L_B^{\text{sub}} = \boldsymbol{\delta}^\top \mathbf{W}_B \boldsymbol{\delta}/2$. These expressions show that multivariate compromise depends on both curvature matrices and on the direction of $\boldsymbol{\delta}$.

S1.9 An exact exchange inequality for the jointly selected value

For any effective losses, positive external task weights, and any dimension of the focal parameter, let θ^*_{joint} minimize $J = \nu_A L_A + \nu_B L_B$. Comparing J at θ^*_{joint} and at θ^*_A, and using $L_A(\theta^*_{\text{joint}}) \geq L_A(\theta^*_A)$,

$$\nu_B \left[L_B(\theta^*_{\text{joint}}) - L_B(\theta^*_A) \right] \leq -\nu_A \left[L_A(\theta^*_{\text{joint}}) - L_A(\theta^*_A) \right] \leq 0,$$

so $L_B(\theta^*_{\text{joint}}) \leq L_B(\theta^*_A)$ and, symmetrically, $L_A(\theta^*_{\text{joint}}) \leq L_A(\theta^*_B)$. Two consequences follow. First, population-level bidirectional transfer implies joint adequacy under arbitrary tolerances, because each task optimum is then adequate in the other task and the joint value does at least as well there. Second, rearranging the same comparison gives

$$L_A(\theta^*_{\text{joint}}) - L_A(\theta^*_A) \leq \frac{\nu_B}{\nu_A} \left[L_B(\theta^*_A) - L_B(\theta^*_B) \right],$$

so with equal external task weights and a common tolerance, successful A→B transfer alone already implies joint adequacy, and symmetrically for B→A. The quadratic boundary ordering in the main text is the local specialization of these inequalities.

S2 Simulation 1: Exact probabilities and an independence check

For the Gaussian design in Simulation 1, in which all three estimators have the same marginal precision, the pass probabilities are available in closed form. Let Φ denote the standard normal distribution function, $r = \sqrt{2\varepsilon} = 0.20$ the acceptable radius, and $\tau = 0.025$ the common standard error. For the nonnegative discrepancy values used in the simulation,

$$\Pr(\text{joint adequacy}) = \begin{cases} 2\Phi((r - \delta/2)/\tau) - 1, & 0 \leq \delta \leq 2r, \\ 0, & \delta > 2r, \end{cases} \tag{S25}$$

and

$$\Pr(\text{bidirectional transfer}) = \left[\Phi((r - \delta)/\tau) - \Phi((-r - \delta)/\tau) \right]^2. \tag{S26}$$

The square in Equation S26 follows because the two source estimates use independent task data. At $\delta = 0.30$, the two expressions give 95.45% and 1.0×10^{-9}; the first matches the simulated rate, and the second is consistent with the absence of bidirectional transfer passes in 20,000 replications. They also clarify the boundary behavior. At the bidirectional-transfer population boundary $\delta = 0.20$, each direction passes with probability approximately .50, so the bidirectional rate is approximately .25. At the joint boundary $\delta = 0.40$, the intersection of the acceptable regions is a single point and the pass probability is zero.

Simulation 1 uses subset means to match marginal precision. This makes the correlation between the joint estimate and each task-specific estimate .50. Equation S25 depends only on the distribution of the joint estimator, and Equation S26 only on that of the two source estimators, so this correlation cannot change either pass-probability curve; it can affect only how often the two events co-occur. To check that co-occurrence, we repeated the analysis with three independent estimators having the same marginal standard error. For each of 20,000 replications, we independently drew

$$\hat{\theta}_{\text{joint}} \sim N(0, \tau^2), \qquad \hat{\theta}_A \sim N(-\delta/2, \tau^2), \qquad \hat{\theta}_B \sim N(\delta/2, \tau^2),$$

and applied the same joint and bidirectional transfer criteria. At $\delta = 0.20$, joint adequacy without transfer occurred in 74.98% of the independent replications and 75.20% of the original

paired replications. The rates were nearly identical, indicating that subset-induced dependence does not account for the reported separation. The two pass probabilities alone determine the probability of joint adequacy without transfer to within 0.14 percentage points at every discrepancy in the grid, whatever the dependence between the estimators. At $\delta = 0.20$, it lies between 74.994% and 75.000%, so the difference between the two simulated rates is Monte Carlo error.

S3 Simulation 2A: Calibration and finite-sample analyses

S3.1 Calibration and tolerance

We applied the one-quarter calibration implied by the equal-curvature quadratic result to the stored Simulation 2A losses, without refitting the models or using the psychometric results to choose the factor. As reported in the main text, it exactly aligns the Gaussian population boundary and nearly aligns the psychometric boundary. Figure S1 shows that the curves for the calibrated joint criterion and bidirectional transfer under substitution remain distinct because they evaluate different fitted values. In both panels, the calibrated joint criterion never passes more often than bidirectional transfer and passes less often at every gap where either pass rate is above zero. Calibration aligns a selected population discrepancy boundary; it does not align the full sampling behavior or turn a jointly selected value into a transferred value.

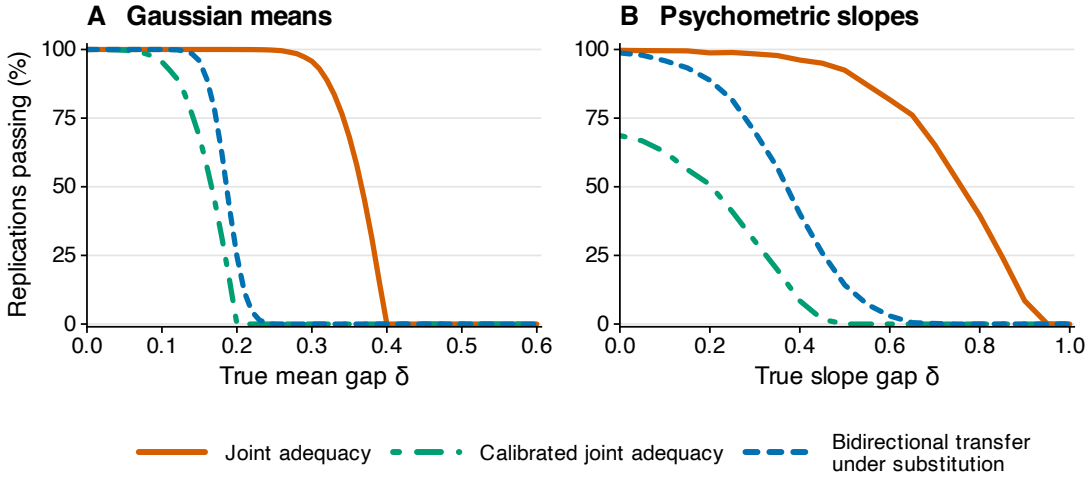


Figure S1: *Aligning a population boundary does not make finite-sample pass rates identical.* Curves show the percentage of replications in which the common-tolerance joint criterion, calibrated joint criterion, and bidirectional transfer under substitution pass as the task-specific discrepancy increases. The joint and transfer criteria use ε; the calibrated joint criterion uses $\varepsilon/4$. This stricter tolerance aligns the Gaussian population boundary with transfer, but the finite-sample curves remain distinct because the joint criteria evaluate the same jointly selected estimate, whereas transfer evaluates estimates selected independently in the source tasks. *(A)* Gaussian means, with 20,000 replications per gap. *(B)* Psychometric slopes, with 2,000 replications per gap and 720 training responses per task.

The ordering is stable across the tolerances examined. When ε was halved, left unchanged, or doubled, the ratio of the common-tolerance joint boundary to the bidirectional-transfer boundary was 1.95, 1.93, and 1.91, respectively.

S3.2 Probability translation

The probability-scale translation fixes the reference location at $\mu = 0$ and uses the reference curve $s = 2$, $\lambda = .02$. Increasing the slope to the point on the $\varepsilon = 0.007$ mean-KL contour produces a largest across-level probability change of 4.97 percentage points; moving to the corresponding lower-slope point produces 5.24 points. The tolerance is thus anchored by a change of about five percentage points at this reference curve, not by a unique conversion between KL loss and response probability. This fixed-location calculation is a behavioral anchor rather than a second decision criterion. At $\delta = 0.60$, the corresponding maxima are 3.53 percentage points for the population shared-slope curve and 6.63 points for the worse of the two transfer directions after profiling the target locations.

S3.3 Optimization and recovery

For both psychometric simulations, we fitted separate and joint models with R's `optim` function, using L-BFGS-B in log-slope and location coordinates. We bounded slope to $[0.1, 8]$ and each location to $[-1.5, 1.5]$. Separate fits began at $(s, \mu) = (2, 0)$, whereas joint fits began at a shared slope of 2 with both locations at zero. We treated an L-BFGS-B code of zero as convergence. If an initial fit did not converge, we retried separate fits from the Cartesian product of $s \in \{0.5, 1, 2, 4\}$ and $\mu \in \{-0.5, 0, 0.5\}$; for joint fits, we retried the four slope starts with both locations at zero. Among the converged finite fits, we retained the one with the lowest negative log likelihood. Conditional target-location fits used one-dimensional `optimize` with the source slope fixed.

A parameter with optimizer coordinate $z \in [\ell, u]$ was flagged as near a bound when $\min(z-\ell, u-z)/(u-\ell) \leq 10^{-4}$; for slopes, $z = \log s$. Table S1 summarizes fit diagnostics and slope recovery. No generated replication was discarded or replaced. Every likelihood-ratio statistic behind the model-selection results was computed only after the code had verified that the separate-slope fit attained a training negative log likelihood no larger than that of the nested shared-slope fit, to within 10^{-6}; a failure would have stopped the run.

Table S1: *Fit diagnostics and slope recovery.* All entries summarize the primary simulation grid in each column; the additional $\delta = 0.60$ recovery benchmark is reported below. Bias and RMSE entries give the maxima across grid values, separately for Tasks A and B. Monte Carlo entries are the largest 95% Wilson half-width among the primary binary outcomes.

Quantity	Simulation 2A	Simulation 2B
Primary-grid replications	62,000	34,000
Primary-grid replications invoking retry	13	4
Primary-grid separate and joint fits converged	186,000/186,000	102,000/102,000
Primary-grid replications with a near-bound fit	0	0
Maximum absolute slope bias (A/B)	0.0211/0.0415	0.0207/0.0265
Maximum slope RMSE (A/B)	0.1565/0.2441	0.1589/0.3377
Maximum 95% MC half-width	2.19 percentage points	2.19 percentage points

The separate-parameter recovery benchmark at $\delta = 0.60$ used 2,000 additional replications. Location bias was 0.0003 for Task A and 0.0004 for Task B, with RMSEs of 0.063 and 0.055. Slope bias was 0.0125 for Task A and 0.0280 for Task B, with RMSEs of 0.131 and 0.192. When $\delta \neq 0$, the joint slope is a population compromise rather than either generating task slope, so recovery against a task-specific generating value is not an appropriate criterion for the joint estimator. In the focused 10,000-replication analysis, the largest Monte Carlo half-width among the unconditional

binary outcomes was about 1.0 percentage point. At each simulated gap from $\delta = 1.05$ through 1.50, neither the joint criterion nor bidirectional transfer passed in any of the 2,000 replications, so joint adequacy without transfer also occurred in none. Parameter-level boundary distances for Simulation 2A and the finite-validation extension, together with retry counts and recovery summaries, accompany the code.

S3.4 Model selection and sample size

With the lapse rate fixed, the shared-slope and separate-slope models estimate three and four parameters, so the likelihood-ratio test has one degree of freedom, and BIC uses the total number of Bernoulli responses across both tasks ($N = 1{,}440$ in the main design). The focused $\delta = 0.50$ analysis in the main text used AIC. In the same 10,000 experiments, the likelihood-ratio test at $\alpha = .05$ failed to reject the shared model in 35.31%, and BIC selected it in 63.37%. Failure to reject coincided with transfer failure in 20.59% of experiments; BIC selection of the shared model coincided with transfer failure in 48.53%. Changing the selection rule therefore changes how often the shared model is retained, but it does not turn model selection into source-to-target transfer. Table S2 gives the complete AIC cross-tabulation.

Table S2: *AIC model selection versus bidirectional transfer at* $\delta = 0.50$. Entries are counts and percentages of 10,000 experiments ($\varepsilon = 0.007$). The joint criterion passed in 91.71% of the same experiments.

AIC decision	Transfer passed	Transfer failed
Shared slope selected	1,077 (10.77%)	702 (7.02%)
Separate slopes selected	407 (4.07%)	7,814 (78.14%)

Sample size can move disagreement in either direction. Each condition used 1,500 experiments with $\varepsilon = 0.007$. At $\delta = 0.60$, the event that AIC selected the shared model while transfer failed fell from 23.27% with 360 training responses per task to 0.07% with 1,440. At $\delta = 0.20$, the converse event, in which AIC selected separate slopes while transfer passed, rose from 7.20% to 44.87%. More data sharpen both decisions, but toward the different population questions defined by model selection and transfer.

S4 Simulation 2B: Design selection and slope reversal

To make the directional contrast visible while holding the model family and trial count fixed, we sought a fourfold profile-curvature contrast. Locally, such a contrast predicts an approximately twofold difference in directional discrepancy boundaries because these boundaries scale as $1/\sqrt{w_T}$. We selected the candidate pair using only reference-curve curvatures at $s = 2$, $\mu = 0$; transfer excess losses and pass outcomes did not enter the selection rule.

Table S3 gives all four candidate grids. The final column reports $w_{[-2,2]}/w_{\text{grid}}$, the broad-grid curvature divided by the current row's curvature. Across all six higher-to-lower-curvature pairs, the $[-2, 2]$ and $[-0.4, 0.4]$ pair gave the ratio closest to four, 4.217; the next closest ratio was 4.509 for $[-1.6, 1.6]$ paired with $[-0.4, 0.4]$.

Because the psychometric loss is nonlinear, the reported population boundaries use exact numerical crossings rather than treating the reference curvature ratio as constant across slope gaps. Reversing the assignment of generating slopes preserved the ordering: B→A transfer remained

Table S3: *Candidate stimulus-grid profile curvatures.* Curvatures are evaluated at the reference configuration used for design selection.

Nine-level grid	Profile curvature	$w_{[-2,2]}/w_{\text{grid}}$
$[-2, 2]$, step 0.5	0.059932	1.000
$[-1.6, 1.6]$, step 0.4	0.064069	0.935
$[-0.6, 0.6]$, step 0.15	0.027809	2.155
$[-0.4, 0.4]$, step 0.1	0.014210	4.217

adequate through $\delta = 0.499$, A→B through $\delta = 0.991$, and the joint criterion passed through $\delta = 1.384$. Task A therefore remained the stricter target.

S5 Nuisance compensation under a one-sided design

To examine the nuisance-profile result in Equation S22 within the psychometric family, we retained the slope construction used in Simulations 2A and 2B:

$$s_A = 2 - \delta/2, \qquad s_B = 2 + \delta/2,$$

and examined A→B transfer on a deliberately one-sided Task B grid with nine levels from 0.2 to 1.8, $\lambda = .02$, $\mu_B = 0.15$, and $\varepsilon = 0.007$. In the fixed-location condition, the target location was held at its population value, $\mu_B = 0.15$. In the profiled condition, the target location was re-optimized for each transferred source slope.

At the reference configuration $s = 2$, $\mu_B = 0.15$, the one-sided grid gives strong slope–location coupling, $\gamma_B = -0.7884$. Profiling the target location widened the A→B discrepancy boundary from 0.488 to 0.850, a factor of 1.74; the local profile-curvature prediction was 1.63. The fixed-location condition treats the target location as known, providing a reference with no nuisance compensation. The wider boundary under profiling shows how the declared nuisance policy changes the transfer claim; this analysis is separate from the Simulation 2B stimulus-grid selection.

S6 Finite-data evaluation

S6.1 A known reference

The primary simulations score each fitted predictor by its exact data-generating excess loss. Their pass rates include training-estimation variation but no finite-validation noise. To isolate that second layer, we generated new training experiments with the Simulation 2A design (720 responses per task), held each fitted predictor fixed, and scored independent additional validation responses. This is not a fixed-budget train–validation comparison.

For each response, we computed the log-score difference between the known generating probability and the fitted predictor. We averaged these differences within each stimulus level and then across the nine levels. Conditional on the fitted predictor, the level-balanced empirical mean log-score difference is an unbiased estimator of its exact mean KL excess loss; realized values can be negative and were not truncated. We used the same validation responses for all predictors within a task. Maxima and tolerance-based decisions are nonlinear, so the finite-validation and exact-loss pass rates need not match.

Validation draws were independent of training. Within each task and training replication, validation sizes were nested at 180, 720, and 2,880 responses, corresponding to 20, 80, and 320 responses at each of the nine stimulus levels. The 180- and 720-response samples were prefixes of the 2,880-response validation draw within each level. At each of five slope gaps, $\delta \in \{0, 0.40, 0.50, 0.60, 0.90\}$, we compared finite-validation and exact-loss decisions for the same fitted parameter estimates across 10,000 training experiments. Disagreement measures validation noise, not error in a test of true parameter equality.

Figure S2 shows that disagreement decreases as validation information increases and is largest when fitted predictors often lie near a tolerance boundary. At $\delta = 0.60$, increasing validation size from 180 to 2,880 responses per task reduced disagreement from 45.82% to 17.79% for the joint criterion, from 14.30% to 3.50% for bidirectional transfer under substitution, and from 53.02% to 20.62% for joint adequacy without transfer.

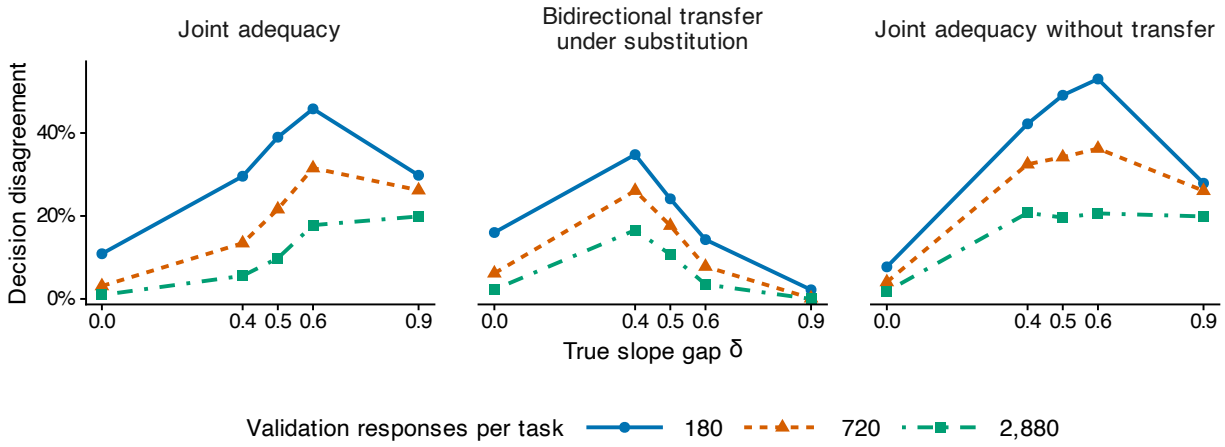


Figure S2: *Decision disagreement falls as validation information increases.* Each panel shows the percentage of 10,000 training experiments in which a decision based on validation data generated independently of the training sample differs from the exact population-loss decision for the same fitted parameter estimates. From left to right, the panels show the joint criterion, bidirectional transfer under substitution, and joint adequacy without transfer. Blue circles, orange triangles, and green squares use 180, 720, and 2,880 validation responses per task, respectively (20, 80, and 320 per stimulus level). Training follows the Simulation 2A design with 720 responses per task and $\varepsilon = 0.007$.

This known-reference extension isolates scoring uncertainty for a fixed predictor; it is not an operational test for an unknown target distribution. The next subsection considers that case.

S6.2 Unknown population optima

In applications, the target's population optimum is unknown, so excess loss cannot be computed directly. A natural substitute holds the source focal value c fixed, fits the permitted target nuisance parameters, fits an unrestricted target model to the same target fitting data, and compares the two fitted models on held-out target observations. Let $R_B(\theta, \phi)$ denote the unprofiled population loss of the target model per observation, so that $L_B(c) = \min_\phi R_B(c, \phi)$ and $L_B(\theta_B^*) = \min_{\theta,\phi} R_B(\theta, \phi)$ on the same scale. Conditional on the fitted nuisance value $\hat{\phi}_S$ of the transfer model and the fitted values $(\hat{\theta}_B, \hat{\phi}_F)$ of the unrestricted model, the expected held-out difference is

$$g_B = R_B(c, \hat{\phi}_S) - R_B(\hat{\theta}_B, \hat{\phi}_F) = \Delta_B(c) + e_S - e_F, \tag{S27}$$

where $\Delta_B(c) = L_B(c) - L_B(\theta_B^*)$ is the population excess loss and

$$e_S = R_B(c, \hat{\phi}_S) - L_B(c) \geq 0, \qquad e_F = R_B(\hat{\theta}_B, \hat{\phi}_F) - L_B(\theta_B^*) \geq 0$$

are the estimation costs of the two fitted models. The held-out difference therefore estimates g_B, not $\Delta_B(c)$. Equation S27 is exact, and the sign and size of the gap depend on $\mathbb{E}(e_S - e_F)$, which is not in general determined by parameter counts. In a correctly specified regular likelihood model with fixed parameter dimension, maximum-likelihood fitting to n target observations, and a transferred value close to the target optimum, $c - \theta_B^* = O(n^{-1/2})$, the expected estimation cost of each fitted model is approximately $k/(2n)$ on the mean negative-log-likelihood scale, where k is the number of estimated parameters; this is the approximation behind the Akaike information criterion. Because the unrestricted model estimates one additional parameter for each focal dimension, under these conditions

$$\mathbb{E}(g_B) \approx \Delta_B(c) - \frac{K}{2n},$$

where K is the dimension of the focal parameter, and the held-out comparison understates the excess loss, which can make transfer look more adequate than it is. If half of the 720 responses in a Simulation 2 task were used for fitting, for example, this local approximation implies a shortfall of about 0.0014 nats per trial for a scalar focal parameter, one fifth of the Simulation 2 tolerance.

In the Gaussian setting of Simulation 1, the population comparison can be made directly from the data. With unit variance and independent task samples of sizes n_A and n_B, the difference between the task means satisfies

$$\hat{\delta} = \bar{Y}_B - \bar{Y}_A \sim N\left(\delta,\ \frac{1}{n_A} + \frac{1}{n_B}\right).$$

On the mean-loss scale of Simulation 1, the population excess loss is $\delta^2/2$ for transfer in either direction and $\delta^2/8$ in each task for the equal-weight joint optimum (Equation S7 with $w_A = w_B = 1$). Both are increasing functions of $|\delta|$, so a confidence interval for δ maps directly onto both losses. Because both loss intervals are derived from the same interval for δ, they hold simultaneously with at least the nominal coverage; the coverage can be conservative because squaring is not one-to-one. Suppose that $\hat{\delta} = 0.30$ with $n_A = n_B = 1{,}600$. The standard error is 0.0354, and the 95% interval for δ is $[0.231, 0.369]$. The corresponding interval for the transfer loss, $[0.0266, 0.0682]$, lies entirely above $\varepsilon = 0.02$, whereas the interval for the joint loss, $[0.0067, 0.0170]$, lies entirely below it. These data therefore support the population conclusion that joint adequacy holds while bidirectional transfer fails, although neither population optimum is known. The intervals concern the losses of the population-optimal source and joint values, not the excess losses of the particular predictors fitted to these samples. Had either loss interval contained ε, the corresponding conclusion would have been inconclusive. If the interval for δ contains zero, the lower limit of both loss intervals is zero.

For models without a closed-form loss surface, the target excess loss must itself be estimated, and the uncertainty analysis should match the claim at issue: a particular fitted source value, the underlying population value, or the behavior of the fitting procedure over repeated samples. In every case, target outcomes may inform permitted nuisance parameters and the benchmark, but not the source focal value. These examples illustrate how finite-data evaluation depends on the estimand and the evaluation design; they are not a generally calibrated inferential procedure for arbitrary cognitive models.